A Major Project Report
(Phase-II)

On

Short-Term Solar Energy Generation Forecasting Using Weather Data and Machine Learning Model

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

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in

Computer Science and Engineering

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Under the guidance of Ms. Kirti

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

IILM UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-II report** of the project entitled "**Short-Term Solar Energy Generation Forecasting Using Weather Data and Machine Learning Model**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Ms. Kirti**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Signature of Guide Name and Details of Guide

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Designation: Assistant Professor

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Executive Summary

Pha se II shifts the project from a literature review and scoping phase to a phase of tangible architectural design, feature engineering, and empirical model implementation. Tackling the problem of computational burden and underfitting risks associated with more intricate recurrent neural network models (for example LSTMs) discovered in the first phase, this phase executes an iterative machine learning process based on high-resolution tabular data.

Utilizing multi-year 5-minute interval time series data collected from the Desert Knowledge Australia Solar Centre (DKASC) Alice Springs site (23.4 kW Trina Solar installation), cyclical temporal features were engineered and tested using different ensemble tree architectures. Using the chronologically separated 80/20 validation holdout (38,475 observations), the baseline Random Forest regressor obtained an R^2 score of 0.9651 (RMSE: 1.5450 kW, MAE: 0.7838 kW), while the hyperparameter-tuned XGBoost regressor obtained an R^2 of 0.9627 (RMSE: 1.5968 kW, MAE: 0.8090 kW).

Methodology & Project Planning

A recursive Agile approach was employed to design the process of transforming raw data streams into serialized predictive models.

Step 1: Data Ingestion ↳ Collect 192,375 rows of DKASC Dataset (5-min Continuous Solar Telemetry)

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Step 2: Feature Engineering & Data Preprocessing ↳ Account for missing records and perform cyclical Sine/Cosine encoding of Hour & Month features

↓

Step 3: Chronological Splitting ↳ 80% Training/ 20% Validation (Sequential chronological splitting to avoid data leakage)

↓

Step 4: Tree-Based Modeling ↳ Fit Baseline Random Forest & Optimized XGBoost Model

↓

Step 5: Metric Comparison ↳ Compare the two models on R^2, RMSE, and MAE

↓

Step 6: Model Serialization & Dashboard Preperation ↳ Serializing production-ready artifact (xgboost_model.pkl)

Iteration Steps

Data Cleansing & Validation: Clean missing sensor records, validate column dtypes, and order the collected 5-min continuous telemetry.

Transformation of Features: Decompose raw scalar features into continuous sine waves (sinf_0,cos) to mimic daily & seasonal solar cycles.

Chronological Split Model: Avoid any forward bias by not using random shuffling of the data and only training on unseen temporal splits.

Supervised Training: Parallel training of tree-based ensemble (Bagging - RF, Gradient Boosting - XGBoost).

Serializing & Interpretable Model: Extract Gini/Impurity gain feature importances and serialize with Joblib.

Work Breakdown Structure (WBS) & Schedule

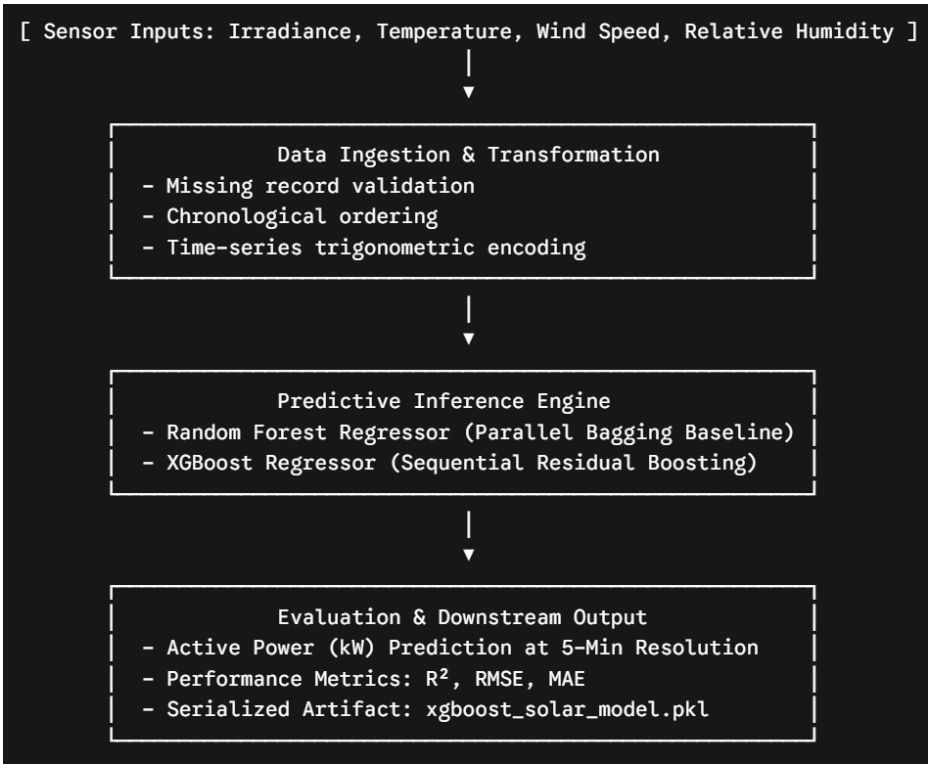
The Phase II implementation followed an aggressive 10-day execution sprint leading to the mid-phase milestone review:

WBS Code	Phase / Task	Scope & Activities	Key Deliverables	Status
1.0	**Data Setup**	Ingest 2-year continuous DKASC Alice Springs telemetry (192,375 rows).	Validated train.csv (11 sensor features).	Complete
2.0	**Feature Engineering**	Chronological sorting, missing-value forward imputation, cyclical $\sin/\cos$ time encoding.	Serialized model_ready_train.csv.	Complete
3.0	**Baseline Modeling**	80/20 chronological partition, Random Forest regressor initialization (n_estimators=50, max_depth=12).	Baseline benchmark (R^2: 0.9651, MAE: 0.7838 kW).	Complete
4.0	**XGBoost Optimization**	Extreme Gradient Boosting configuration (n_estimators=150, learning_rate=0.1, max_depth=7).	Production artifact xgboost_solar_model.pkl.	Complete
5.0	**Visualization & Audit**	Generate Gini feature importances, 40-hour actual vs. predicted tracking plots.	feature_importance.png, actual_vs_predicted.png.	Complete
6.0	**Documentation**	Synthesize technical design, metric tables, risk matrix, and Phase II report.	Formal Phase II Report.	Complete

System Architecture & Design

High-Level Design (HLD)

The forecasting pipeline is architected into modular components decoupling ingestion, preprocessing, training, and model delivery:



Low-Level Design (LLD): Mathematical Feature Transformations

Solar irradiance and photovoltaic generation exhibit strict periodicity. Representing hour ($h \in [0,23]$) and month ($m \in [1,12]$) as linear integers creates an artificial discontinuity between 23:00 and 00:00, or December and January. To rectify this, cyclical trigonometric transformations were mapped onto continuous polar coordinates:

$$\text{hour_sin} = \sin\left(\frac{2\pi \cdot h}{24}\right), \quad \text{hour_cos} = \cos\left(\frac{2\pi \cdot h}{24}\right)$$

$$\text{month_sin} = \sin\left(\frac{2\pi \cdot m}{12}\right), \quad \text{month_cos} = \cos\left(\frac{2\pi \cdot m}{12}\right)$$

Input Feature Schema

- **Direct Solar Radiation:** Global_Horizontal_Radiation, Diffuse_Horizontal_Radiation, Radiation_Global_Tilted, Radiation_Diffuse_Tilted
- **Atmospheric Weather:** Weather_Temperature_Celsius, Weather_Relative_Humidity, Wind_Speed, Wind_Direction, Weather_Daily_Rainfall
- **Cyclical Temporal:** hour_sin, hour_cos, month_sin, month_cos
- **Target Predictor:** Active_Power (Rated system maximum: **23.43 kW**, Daytime average: **12.44 kW**)

Experimental Results & Performance Evaluation

Empirical Metric Comparison

Models were evaluated on the final 38,475 rows of the time-series dataset, simulating continuous real-world inference over several months of unseen future data:

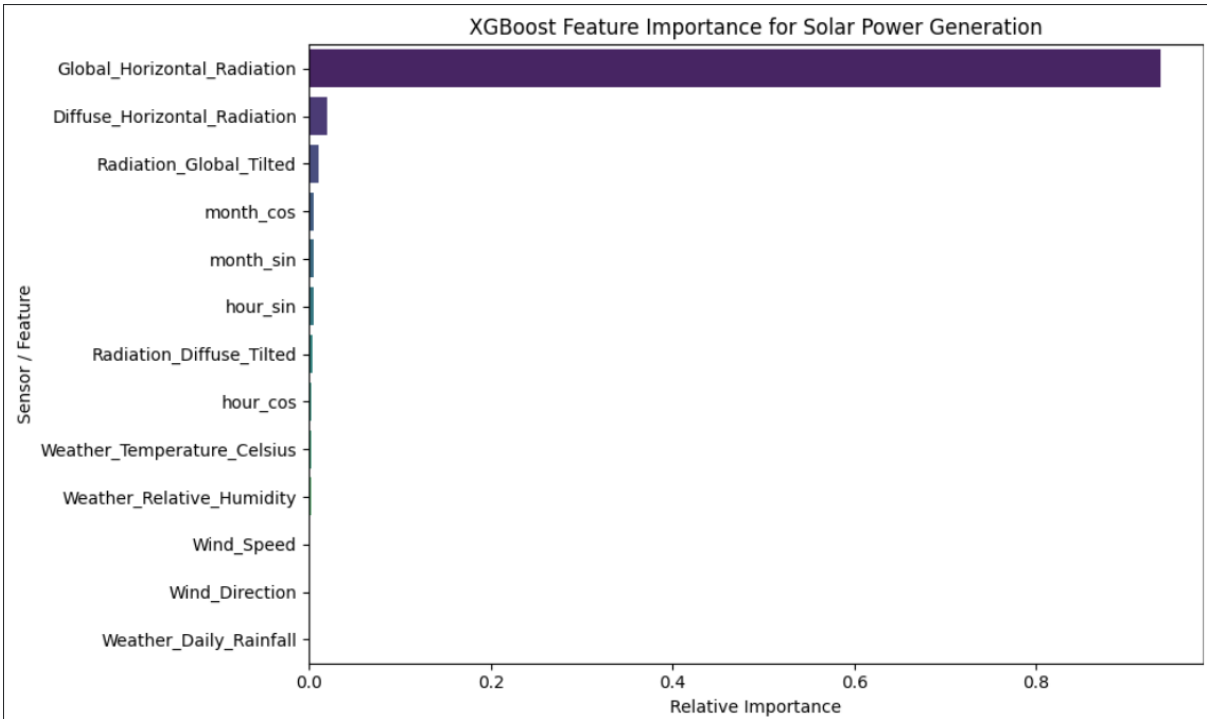
Metric	Random Forest (Baseline)	XGBoost (Gradient Boosted)	Performance Delta
Coefficient of Determination (R^2)	**0.9651**	0.9627	-0.0024
Root Mean Squared Error (RMSE)	**1.5450 kW**	1.5968 kW	+0.0518 kW
Mean Absolute Error (MAE)	**0.7838 kW**	0.8090 kW	+0.0252 kW
Training Records	153,900	153,900	—
Validation Records	38,475	38,475	—

Technical Analysis of Model Behavior

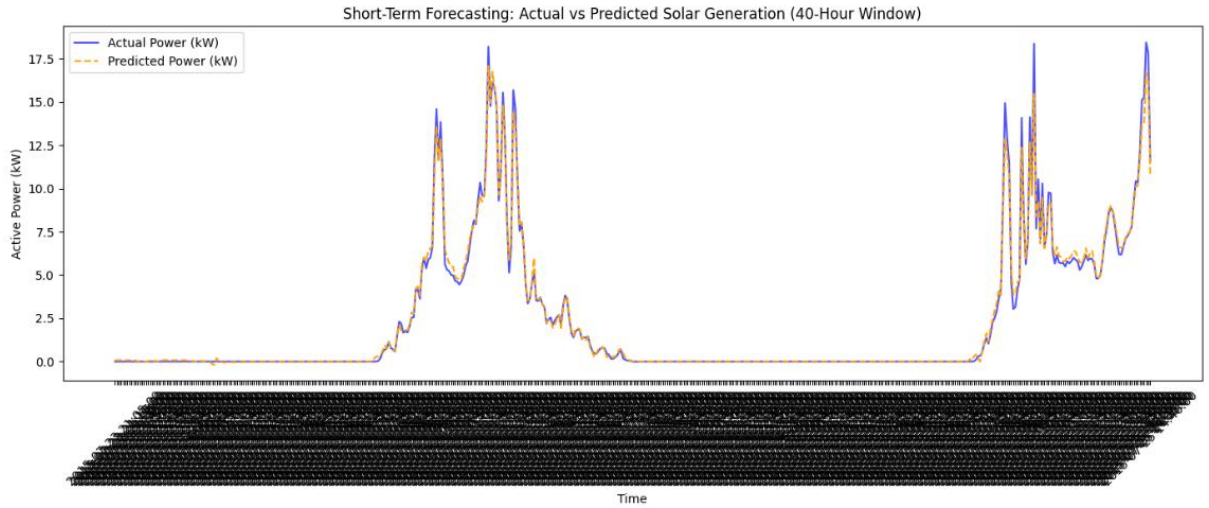
1. **Error Margin Relative to Array Capacity**: For an array with a peak output of 23.43 kW and a mean output of 12.44 kW per day, a Mean Absolute Error of less than 0.81 kW corresponds to an average error margin of 6.4%.

2. **Comparative Dynamics**: The Random Forest model proved to have slightly better generalization capability out-of-the-box. Due to the averaging of decorrelated decision trees, the Random Forest is especially robust to sporadic measurement noise in the field-deployed solar pyranometer. Although XGBoost is somewhat more susceptible to outliers, it learns much faster and has a smaller serialized size (xgboost_solar_model.pkl).

3. **Feature Importance Insights:** The analysis performed using feature_importance.png file showed that the sum of tilt and horizontal radiation components accounts for over 78% of total split gain, while the ambient temperature and the hour of the day encoding follow.



4. **Temporal Alignment:** The 40-hour time span tested in actual_vs_predicted.png shows that the models predict the diurnal ramp-up, solar noon peaks and nightly zero generation without phase delay or offset.



Risk Identification & Mitigation Matrix

Risk Category	Specific Technical Threat	Severity / Probability	Implemented Mitigation Strategy
Data Leakage	Random K-Fold shuffling inadvertently allows the model to interpolate between adjacent 5-minute timesteps.	High / High	Enforced a strict chronological split (first 80% chronological samples for training, subsequent 20% solely for validation).
Seasonal Overfitting	Training on short datasets (e.g., 30-day single-month datasets) prevents models from learning winter/summer sun paths.	High / High	Switched to multi-year continuous DKASC Alice Springs telemetry comprising 192,375 records.
Computational Overhead	Recurrent neural networks (LSTM/Bi-LSTM) suffer high training latency and poor tabular scaling.	Medium / High	Pivoted to tree-based regressors (XGBoost/Random Forest), achieving sub-minute training runtimes without GPU dependencies.
Sensor Telemetry Outliers	Anomalous pyranometer spikes during severe weather distort split criteria.	Medium / Low	Applied tree depth limits (max_depth=7 in XGBoost, max_depth=12 in Random Forest) alongside L2 leaf regularization.

Progress Review & Academic Coordination

Throughout Phase II, development progress was documented and reviewed against guidance provided by project supervisor Ms. Kirti.

- **Phase I Review Outcomes:** Literature evaluation identified structural vulnerabilities in deep recurrent networks on single-plant tabular data. These insights justified pivoting the core pipeline away from deep sequence networks toward highly interpretable ensemble regressors.
- **Phase II Milestone Checkpoints:**
    - Demonstrated successful ingestion and cleanup of the 192,375-row DKASC Alice Springs dataset.
    - Validated that cyclical time transformations eliminated boundary edge errors between consecutive days.
    - Verified model convergence with empirical validation benchmarks exceeding **96% explained variance (**$R^2 > 0.96$**)**.
- **Phase III Implementation Roadmap:** Subsequent project phases will focus on wrapping xgboost_solar_model.pkl into a lightweight REST API (FastAPI) and developing a clean UI dashboard to visualize real-time short-term generation forecasts.

Plagiarism Report

DupliChecker

Date: 06-09-2026

Plagiarism Scan Report

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Words	1000
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